



HaBB DiveSync

User manual — synchronising underwater video, USBL and CTD data for
habitat mapping

Version 1.0

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1. Introduction and overall workflow

HaBB DiveSync is a tool for marine geologists doing habitat mapping with ROVs, submersibles or underwater cameras. Its purpose is to synchronise the video with the USBL positioning file and with CTD data, and to extract high-quality frames ready for use in ArcGIS or annotation software.

Typical pipeline:

1. Load video, USBL and CTD in tab 1.
2. Let the tool detect every available video timestamp and pick the right one.
3. Map USBL columns (lat, lon, depth, time) — tab 2.
4. Map CTD columns (time or depth, parameters) — tab 3.
5. Go to tab 5 (Sync & Extraction): align video↔USBL via touchdown, then align CTD via drag/auto-fix.
6. Set extraction range and quality parameters, run extraction.

Output: PNG/JPG/TIFF frames, CSV with all coordinates and CTD parameters per frame, optionally a shapefile.

2. Installation and first launch

Quick start (executable):

Double-click `HaBB_DiveSync.exe` or `avvia_rov_Tool.bat`. The Logos folder must sit next to the executable to display the logo correctly in the About screen.

Run from Python sources:

`pip install -r requirements.txt`, then `python rov_sync_tool.py`

Optional but recommended dependencies:

- **pymediainfo** — extracts container metadata timestamps. Without it the tool falls back to text sidecars, MediaInfo CLI or the internal binary parser.
- **MediaInfo CLI** — useful as fallback. On Windows install it from mediaarea.net and keep `MediaInfo.exe` in PATH or in `C:\Program Files\MediaInfo\`.
- **geopandas** — enables shapefile export.
- **OpenCV with CUDA** — accelerates video decoding on NVIDIA GPUs.

3. Tab 1 — Files and video timestamp

Load the three files of the dive: video, USBL, CTD. The trickiest part is the video start timestamp — without it nothing else can be synchronised.

3.1 — Robust timestamp detection

HaBB DiveSync uses four cascading strategies to detect the video start timestamp:

- ① **Sidecar .mov.txt / .mp4.txt** — text file produced by MediaInfo CLI alongside the video, containing Encoded date, Tagged date and Time code of first frame.
- ② **pymediainfo** (Python library) — scans every attribute of General and Other tracks.
- ③ **MediaInfo CLI** via subprocess — invoked when `pymediainfo` is missing.
- ④ **Built-in binary parser** for the ISOBMFF tmcd box (MOV/MP4).

3.2 — All-timestamps picker

Below the ■ Detect timestamp button there is the «**All available timestamps**» section. It lists every timestamp found as a colour-coded radio button:

- • **green** = high reliability (Encoded date, Time code, Tagged date)
- • **yellow** = medium reliability (filename pattern, binary parser)
- • **red** = low reliability (filesystem mtime/ctime/atime, pymediainfo file_creation_date)

Pick the right entry and press ■ **Use selected as video start**. The value is propagated automatically to the «Corrected video start timestamp» field in the Sync tab.

If a QuickTime timecode only provides HH:MM:SS without a date, the tool combines it automatically with the USBL date (when USBL is loaded) to obtain a complete timestamp.

3.3 — Filename pattern

For files with a timestamp in the name (e.g. 20221012_085134_DIVE.mov), enter a pattern like YYYYMMDD_HHMMSS_* in the «Filename timestamp pattern» field.

4. Tab 2 — USBL mapping

Tell the tool how to interpret the USBL file.

4.1 — Timestamp

Three modes: **unified** (e.g. ISO 8601 in one column), **split** (date and time in separate columns), **unix** (epoch seconds).

4.2 — Coordinates

Five formats are supported, in this order: **NMEA**, **DD**, **DM**, **DMS**, **UTM**.

- • **NMEA** — DDMM.mmmm,N format typical of GPGGA / PSIMSSB strings.
- • **DD** — decimal degrees (e.g. 28.31633).
- • **DM** — degrees and decimal minutes (e.g. 28°18.98').
- • **DMS** — degrees, minutes, seconds (e.g. 28°18'58.6").
- • **UTM** — UTM coordinates with zone and hemisphere.

4.3 — Validation

Press «Validate USBL» to read the whole file, build the unix_ts column, convert coordinates and show the first row. If you get out-of-range warnings, double-check the selected coordinate format.

5. Tab 3 — CTD mapping

Decide how to pair the CTD with the video frames.

5.1 — Sync mode

- • **By Time** — the CTD is on the ROV or shares a clock. The tool finds the CTD record with the closest time to each frame.
- • **By Depth** — the CTD is independent (e.g. cast performed before the ROV dive). The tool finds the CTD record with the closest depth to the USBL depth of each frame.

5.2 — Window selection (By Depth only)

If the CTD contains both downcast and upcast, the depth profile lets you graphically select the relevant portion (drag on the plot, or use «Downcast» / «Upcast» / «All» buttons).

6. Tab 4 — Custom CSV columns

Add fixed columns to the output CSV: dive name, ROV code, operator, site, anything you need for GIS archiving.

7. Tab 5 — Sync & Extraction

The heart of the program: where the actual synchronisation happens.

7.1 — Video slider with absolute time

Below «Posizione video (s)» there is the «**Time UTC**» field. It shows in real time the UTC timestamp matching the current slider position. You can also type a time inside it (e.g. 2020-10-25T11:55:00 or 11:55:00) and press Enter or «**Go**» to jump straight to that frame without dragging the slider — vital for multi-GB videos.

7.2 — Touchdown sync (Video↔USBL)

1. **Move the slider** to the frame where the ROV touches the bottom.
2. **Click on the USBL plot** at the actual touchdown point.
3. **Press Synchronise with these two points**: the tool computes the video↔USBL delay and writes it to «Corrected video start timestamp».

The **Synchronise** / **Reset** / **Update preview** buttons sit between the Video box and the USBL plot, so they act right under the two points you've just set.

7.3 — CTD alignment: the interactive plot

Below the USBL plot there is the **CTD ↔ USBL depth overlay**: two superimposed curves, USBL in blue (fixed), CTD in red (draggable along the X axis).

Three interchangeable, cumulative ways to align the CTD:

- **Auto-align (depth match)** — computes the shift that brings the CTD depth of the first frame to match the USBL depth.
- **Manual drag** — click on the plot and drag the red curve until it overlaps the blue one. The curve moves in real time.
- **Number entry** — type a value into «CTD shift (s)» in the *Manual correction* box and press Enter: the red curve shifts by that many seconds.

The «CTD shift (s)» field is the single source of truth: drag, auto-fix and manual entry all act on that field. You always see it updated live.

Buttons next to the field:

- **Apply** — bakes the current shift into the validated CTD series and resets the field to 0, creating a new baseline.
- **Reset** — sets the field to 0 without modifying the validated df (cancels the preview).

During extraction the current value of «CTD shift (s)» is applied even if you didn't press Apply: baking is optional.

7.4 — Depth tolerance + extraction lock

The field «**Max |Δdepth CTD–USBL| (m)**» sets the maximum tolerance. Below it, a status label tells you in real time:

- •  **depth CTD≈USBL** — within tolerance, OK.
- •  **depth CTD ≠ USBL → extraction blocked** — out of tolerance: extraction will ask for explicit confirmation before proceeding.

7.5 — Double sync (CTD by_time)

In by_time mode you must verify TWO things:

- • CTD time aligns with USBL/video time (small ΔT in the table);
- • CTD depth matches USBL depth (|ΔD| under threshold).

Typical workflow: start with  Auto-align (depth match) for the order-of-magnitude shift, then refine by dragging until the status label turns green.

7.6 — Alignment preview table

Ten sample frames with: index, video time, USBL time, ΔT USBL, USBL depth, CTD time/depth, ΔT CTD (or Δdepth in by_depth), CTD depth, |ΔD vs USBL|. Rows turn red when tolerances are exceeded.

7.7 — Extraction range (video portion)

Two rows of fields: **From / To in seconds** and **From / To as UTC**. They update each other: type a UTC time and the seconds change automatically, and vice versa. The « From current pos.» / « To current pos.» buttons grab the slider's current position.

8. Frame extraction and outputs

Parameters:

- • **Every N seconds** — interval between consecutive frames.
- • **Association window (s)** — maximum tolerance to accept a USBL/CTD match.
- • **Format** — PNG (lossless), JPEG (compressed), TIFF.
- • **Quality controls** — blur/dark/bright thresholds to flag bad frames.
- • **Overlay** — optional text on frames (time, depth, lat/lon, dive, CTD parameters).

Outputs:

- • `imageNNNN.png` — extracted frames
- • `{dive_name}.csv` — CSV with frame, time, lat, lon, USBL/CTD depth, CTD parameters, blur/brightness, warnings, custom columns
- • `{dive_name}.shp` — shapefile (if geopandas is installed)
- • `{dive_name}_session.json` — extraction session log

9. Troubleshooting

Issue: the video timestamp shows only file_creation_date (today's date).

Cause: your MOV was (re)encoded by FFmpeg/Lavf and contains no encoded_date. pymediainfo reports file_creation_date, which is just the file creation time on disk.

Fix: open the «All available timestamps» section and look for a green «*sidecar — Time code of first frame*» entry. If absent, generate a sidecar with `mediainfo "file.mov" > "file.mov.txt"`

and press Detect timestamp again.

Issue: dragging the CTD curve doesn't work.

Cause: CTD not validated, or depth column not mapped.

Fix: go back to tab 3, choose the depth column and press «■ Validate CTD».

Issue: extraction blocked by «depth CTD $\neq$ USBL».

Cause: CTD shift still wrong or CTD belongs to a different dive.

Fix: drag the red curve to overlap the blue one, or raise «Max $|\Delta\text{depth}|$ (m)».

Issue: the tool is slow when seeking far away with the slider.

Cause: multi-GB MOV files require linear decoding to reach a specific point.

Fix: use the «Time UTC» or «Posizione video (s)» fields in the Sync tab to jump directly to a target timestamp / second instead of dragging the slider.

10. Quick reference

Shortcuts:

- • Enter in «Time UTC» or «Posizione video (s)» → jump to that point.
- • Enter in «From (UTC)» / «To (UTC)» → recompute «From (s)» / «To (s)».
- • Enter in «CTD shift (s)» → redraws the CTD curve with the new shift.
- • Shift+mouse wheel → horizontal scroll inside the current tab.

Minimal pipeline:

Load the 3 files → Detect timestamp + pick from the picker → Map USBL → Validate USBL → Map CTD → Validate CTD → Touchdown video↔USBL sync → ■ Auto-align CTD → fine drag → ■ Apply → Set extraction range (with UTC times) → Start.

Contacts:

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